Conscription and Workplace Adjustment: Reallocation, Hiring, and Long-Run Career Effects

- Conflicts and wars displaces workers from their workplaces.
  Japanese bill in 1945 mobilized 65% of the working population into the battlefields.
- Replacement costs for employers soar during wartimes
  - Internal markets: Younger staff are often abscent due to conscription.
  - External markets: Shortage of labor supply.

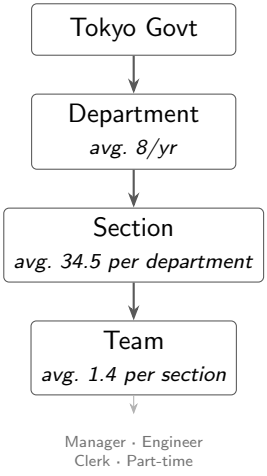
Research Questions

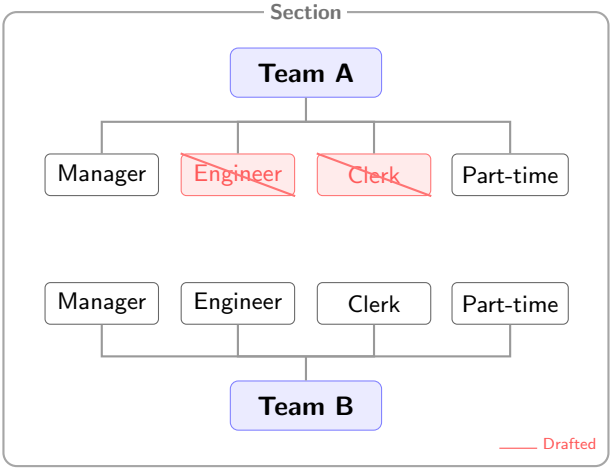
- How are the vacancies filled by reassinging incumbent workers into different roles through different contracts?
- What are the long run effect of such temporarly reassignment on the career trajectories of the colleagues surrounding the draftee?

Identifying who filled draft-induced vacancies

- We study how draft-induced vacancies in a public-sector organization with more than 40,000 employees (e.g., the Tokyo civil service) were filled.
- We compare vacancy-filling patterns across organizationally adjacent sections in response to draft-induced employee exits.
  - The drafting process was undisclosed to the public $\Rightarrow$ drafting is as good as random after conditioning on occupation and section assignment.
- We compare how vacancies are filled through internal transfers versus external hiring.

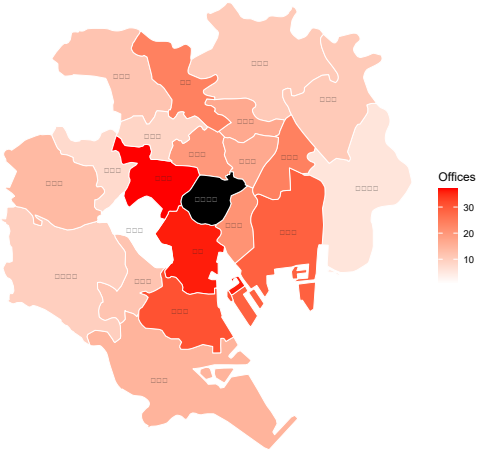
Identification: within-section, within-occupation





Balance test (backup)

Tokyo offices across wards



Empirical strategy: OLS with fixed effects

- For each hiring/transfer margin, we estimate:

$$Y_{okt} = \beta \cdot \text{Draftees}_{okt} + \theta \cdot \text{Workers}_{kt} + \alpha_o + \gamma_k + \varepsilon_{okt}$$

- Y_{okt}: hiring or transfer margin in occupation o, section k, and year t.
- Draftees_{okt}: number of drafted workers.
- Workers_{kt}: number of workers in section k in year t.
- α_o: occupation fixed effects.
- γ_k: section fixed effects.

Drafting and replacement margins

Panel A. Composition of the 2:1 ratio (internal vs. external)

	Internal staff response ≈ **0.41**				*External* ≈ **0.21**
	Total internal	— decomposition — Retention (same section)	Same dept. (cross-section)	Diff. dept. (cross-bureau)	New hires
No. drafted workers	0.4079***	0.2269***	0.1001**	0.0809***	0.2057**
	(0.0761)	(0.0730)	(0.0403)	(0.0154)	(0.0998)
Ratio: internal / external = 0.41 / 0.21 ≈ **2 : 1**					
Observations			7,409		
Year FE			Yes		
Section FE			Yes		
Position FE			Yes		
Log cumulative male baseline			Yes		

Notes: OLS regressions. Unit of observation: destination office × kakari × occupation × year, 1938–1945. Each cell reports the coefficient on the number of drafted male workers in the same destination kakari-occupation-year. "Retention" counts workers who remained in the same ka group (same-section stayers). "Same dept." counts workers from a different ka group within the same normalized kyoku group (cross-section transfers). "Diff. dept." counts workers from a different normalized kyoku group (cross-bureau transfers). "Total internal" is the sum of all three. New hires exclude first observed office-years. All specifications include year, section, and position fixed effects and control for log cumulative male baseline employment. Standard errors clustered by office in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Bilateral mobility: nested organizational distance

Unit: origin office $\times$ origin occupation $\times$ destination office $\times$ destination occupation $\times$ year.

$$E[F_{op,dq,t}] = \exp\{\beta_1 \log N_{op,t-1} + \beta_2 \log N_{dq,t-1} + \beta_3 \log N_{op,t-1} \log N_{dq,t-1} + \beta_4 \text{OccSimilarity}_{od,t-1} + \beta_5 \mathbf{1}\{\text{same kyoku, diff. ka}\} + \beta_6 \mathbf{1}\{\text{diff. kyoku}\} + \alpha_t +$$

- Retention is same office and same occupation; excluded from the flow panel.
- Baseline category: workers change occupation within the same ka $\times$ kyoku.
- **Same kyoku, different ka**: within-department cross-section mobility.
- **Different kyoku**: cross-department mobility. Sample excludes office-years with drafting at either endpoint.

Bilateral mobility: strong nested distance gradient

	(1) Year FE	(2) Occ-pair FE	(3) Origin kyoku FE	(4) Dest. kyoku FE	(5) Origin ka FE	(6) Dest. ka FE
log $N_{o,p}$	0.778***	0.861***	0.773***	0.781***	0.855***	0.763***
	(0.075)	(0.099)	(0.077)	(0.076)	(0.074)	(0.092)
log $N_{d,q}$	0.239***	0.130	0.218***	0.222***	0.163**	0.151**
	(0.060)	(0.099)	(0.062)	(0.064)	(0.072)	(0.066)
log $N_{o,p} \times$ log $N_{d,q}$	0.007	-0.041	0.012	0.010	0.036	0.031
	(0.019)	(0.027)	(0.019)	(0.019)	(0.025)	(0.025)
Occupation similarity	1.788***	1.090***	1.536***	1.404***	1.574***	1.296***
	(0.359)	(0.381)	(0.366)	(0.384)	(0.371)	(0.422)
Same kyoku, diff. ka	-2.334***	-3.519***	-2.324***	-2.375***	-2.361***	-2.408***
	(0.326)	(0.339)	(0.346)	(0.358)	(0.340)	(0.344)
Different kyoku	-4.659***	-5.889***	-4.860***	-4.943***	-4.897***	-5.025***
	(0.271)	(0.272)	(0.298)	(0.287)	(0.290)	(0.290)
Observations	4,445,908	2,684,319	4,406,699	3,590,943	2,778,413	1,420,716
Pseudo R^2	0.333	0.416	0.352	0.357	0.389	0.410

Notes: PPML regressions. Unit is origin office–occupation $\times$ destination office–occupation $\times$ year, 1938–1945. Retention, defined as same office and same occupation, is excluded. Omitted category is same office (ka $\times$ kyoku), different occupation. A literal same-kakari baseline has no positive flow cells in the kakari-covered sample. Standard errors clustered two-way by origin and destination office. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Selection into draft-induced vacancies

	All transfers		Different ka, same kyoku		Different kyoku	
	Any: Draft Vac.	Any: Non-Draft	Diff Ka: Draft	Diff Ka: Non-Draft	Diff Kyoku: Draft	Diff Kyoku: Non-Draft
Decoration rank (0–8; high=better)	-0.0529	0.2466	-15.4652***	0.2863	0.1670	0.2020
	(0.4393)	(0.1728)	(0.0000)	(0.1787)	(0.4425)	(0.2030)
Court rank (0–8; high=better)	-0.0186	-0.1181**	-0.3238	-0.0864	0.0372	-0.1504**
	(0.1355)	(0.0601)	(0.3940)	(0.0667)	(0.1420)	(0.0734)
Female	-1.0508	0.3935*	-18.5681***	0.4252*	-0.6950	0.3173
	(0.8136)	(0.2358)	(0.0000)	(0.2467)	(0.8277)	(0.2850)
Tenure (yrs)	0.1014***	0.0295**	-0.0950**	-0.0209	0.2123***	0.0965***
	(0.0223)	(0.0125)	(0.0414)	(0.0139)	(0.0258)	(0.0145)
Log salary (prior year)	-0.1404	-0.2396***	0.1765	-0.1763*	-0.3365*	-0.3171***
	(0.1625)	(0.0921)	(0.2320)	(0.1012)	(0.1891)	(0.1050)
Year FE			Yes			
Observations			5,523			
Log-likelihood	-3140.5				-5295.7	
AIC	6329.1				10687.5	

Sample restricted to origin office-unit × year cells with at least one transfer to a different ka or kyoku, further restricted to workers with observed prior-year salary. Base category: stayed in same ka and kyoku (required for MNL interpretation). Rank variables are coded 0–8, with higher values indicating higher status.

Trajectory regression: transferees vs peers

	Tenure after	Postwar surv.	Postwar years	Rank gain
Transferred vs. workers who **stayed at donor**	3.7525***	0.1953***	0.6345***	0.1813***
	(0.0911)	(0.0052)	(0.0206)	(0.0071)
Draft-induced vs. **natural-exit** destination	1.5513***	0.0801***	0.4697***	0.0304**
	(0.2350)	(0.0153)	(0.0649)	(0.0137)
Transferred vs. workers **already at destination**	2.2992***	0.1426***	0.4895***	0.1348***
	(0.0868)	(0.0059)	(0.0208)	(0.0060)
Fixed-effects				
Team	Yes	Yes	Yes	Yes
Occupation	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
Observations	Row 1: 91,586	Row 2: 40,060	Row 3: 97,145	

*Clustered SE in parentheses; ***: 0.01, **: 0.05, *: 0.1*

Promotion and bonus salary responses

	No. promoted workers		*No. bonus recipients*	
Model:	(1)	(2)	(3)	(4)
Variables				
No. drafted workers	0.0267*	0.0231*	0.1367**	0.1548***
	(0.0107)	(0.0114)	(0.0529)	(0.0587)
No. drafted × Part-time		0.0154		−0.1854**
		(0.0112)		(0.0610)
Fixed-effects				
Year / Section / Occupation	Yes	Yes		
Section / Occupation			Yes	Yes
Log section size	Yes	Yes	Yes	Yes
Observations	2,770	2,770	576	576

*Clustered SE in parentheses; ***: 0.01, **: 0.05, *: 0.1*

Further analysis

- How were promotions and bonus pay used to transfer or retain incumbent workers?
  Findings: Both were used more in impacted offices. Promotions were used more often together with transfers.
- How did early promotion help transferred workers in their subsequent careers?
  Findings:
  1. Transferees sent to drafted offices outperform those sent to non-drafted offices.
  2. Transferees from remote teams perform worse than transferees from neighboring teams.

Discussion and conclusion

- Workplaces compensated for staff losses jointly through internal and external markets (about 2:1 per drafted employee).
- These results support empirical work on responses to worker exits (Jaeger, Heining, and Lazarus, 2025; the worker-death paper).
  But we show that replacement costs varied with the composition of workers surrounding each affected worker.
- Transferred workers presumably accelerated promotion and contributed to better career trajectories.

References I

Empirical strategy: who is selected for transfer?

For each donor office unit-year, worker i chooses among transfer outcomes $j \in \{\text{stay}, \text{draft vacancy}, \text{non-draft vacancy}\}$:

$$\Pr(y_{it} = j \mid X_{it}) = \frac{\exp\left(X_{it}'\beta_j\right)}{\sum_{\ell \in \mathcal{J}} \exp\left(X_{it}'\beta_\ell\right)}, \qquad X_{it}'\beta_{\text{stay}} = 0.$$

$$X_{it}'\beta_j = \beta_{0j} + \beta_{1j}\text{Decor}_{i,t-1} + \beta_{2j}\text{Court}_{i,t-1} + \beta_{3j}\text{Female}_{it} + \beta_{4j}\text{Tenure}_{it} + \beta_{5j}\log(\text{Salary}_{i,t-1} + 1) + \lambda_{jt}.$$

- Outcomes are estimated separately for leaving at all, within-kyoku transfers, and cross-kyoku transfers.
- Decoration and CourtRank are coded 0–8, with 0 for unobserved status.
- Female is an indicator, Tenure is years since first observed, and λ_{jt} are outcome-specific year fixed effects.

Drafting and replacement margins: geographic benchmark

Panel A. Composition of the 2:1 ratio (internal vs. external)

	Internal staff response ≈ **0.41**				*External* ≈ **0.21**
	Total internal	— decomposition — Retention (same section)	Same dept. (cross-section)	Diff. dept. (cross-bureau)	New hires
No. drafted workers	0.4079***	0.2269***	0.1001**	0.0809***	0.2057**
	(0.0761)	(0.0730)	(0.0403)	(0.0154)	(0.0998)

Ratio: internal / external = 0.41 / 0.21 ≈ **2 : 1**

Panel B. Geographic benchmark for internal replacements

	Total internal	≤2 km	>2 km	Unknown dist.	New hires
No. drafted workers	0.4079***	0.2506***	0.0780	0.0793	0.2057**
	(0.0761)	(0.0860)	(0.0529)	(0.0642)	(0.0998)
Observations			7,409		
Year FE			Yes		
Section FE			Yes		
Position FE			Yes		
Log cumulative male baseline			Yes		

Notes: OLS regressions. Unit of observation: destination office × kakari × occupation × year, 1938–1945. Each cell reports the coefficient on the number of drafted male workers in the same destination kakari-occupation-year. "Retention" counts workers who remained in the same ka group (same-section stayers). "Same dept." counts workers from a different ka group within the same normalized kyoku group (cross-section transfers). "Diff. dept." counts workers from a different normalized kyoku group (cross-bureau transfers). "Total internal" is the sum of all three. Panel B splits internal replacements by great-circle distance between prior-year and current-year office ward centroids. New hires exclude first observed office-years. All specifications include year, section, and position fixed effects and control for log cumulative male baseline employment. Standard errors clustered by office in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Peace-time departures and replacement margins

Tokyo-Shi baseline, 1929–1937

	Internal staff response				*External*
	Total internal	— decomposition — Retention (same section)	Same dept. (cross-section)	Diff. dept. (cross-bureau)	New hires
Natural male departures	-0.0170	0.0125	-0.0220**	-0.0074	-0.0064
	(0.0196)	(0.0152)	(0.0098)	(0.0102)	(0.0108)
Observations			3,950		
Year FE			Yes		
Section FE			Yes		
Position FE			Yes		
Baseline male employment (log)			Yes		

Notes: OLS regressions using Tokyo-Shi directory transitions, 1929–1937. Unit of observation: destination office × kakari × occupation × year. The treatment is the number of male workers in the same office-kakari-occupation cell in $t-1$ who are no longer in that exact cell in year t, excluding draft-coded rows. Outcomes use the same replacement-margin definitions as the wartime draft table: retention within the same ka group, transfers from a different ka group within the same normalized kyoku group, transfers from a different normalized kyoku group, and new hires excluding first observed office-years. All specifications include year, section, and position fixed effects and control for log cumulative male baseline employment. Standard errors clustered by office in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Balance test of treatment and control

Outcome (pre-war)	Extensive coef.	p-value	Intensive coef.	p-value
No. new hires	-2.05	0.094	-0.0987	0.446
	(1.15)		(0.1263)	
No. internal transfers	40.5	0.533	1.1205	0.025
	(63.5)		(0.4533)	
No. promoted	0.464	0.055	0.0132	0.146
	(0.224)		(0.0087)	
Female share	-0.0014	0.859	-0.0002	0.152
	(0.0080)		(0.0001)	
Sample size (obs.)	571 (outcome-specific in full table)			
Controls	FE: Section + position + year; + log team-cell size			

Back to identification slide

Research question idea: Tokyo bombing shock

- How did the 1945 air-raid displace managers from teams?
- In 1947, Tokyo removed the bar for women to take managerial positions.
  $\Rightarrow$ Number of female managers increased from 10 to 100 in 1948.
- Did women managers get assigned to teams with a displaced manager?
- Did women managers retain/promote female employees more than male employees?